

Q1 - 24 June - Shift 1

From the top of a tower, a ball is thrown vertically upward which reaches the ground in 6 s.

A second ball thrown vertically downward from the same position with the same speed reaches the ground in 1.5 s. A third ball released, from the rest from the same location, will reach the ground in _____ s.

Space for your notes:

Q2 - 24 June - Shift 2

An object of mass 5 kg is thrown vertically upwards from the ground. The air resistance produces a constant retarding force of 10 N throughout the motion. The ratio of time of ascent to the time of descent will be equal to :
[Use $g = 10 \text{ ms}^{-2}$]

Space for your notes:

(A) 1 : 1 (B) $\sqrt{2} : \sqrt{3}$

(C) $\sqrt{3} : \sqrt{2}$ (D) 2 : 3

Q3 - 25 June - Shift 2

Questions

MathonGo

Two buses P and Q start from a point at the same time and move in a straight line and their positions are represented by $X_P(t) = \alpha t + \beta t^2$ and $X_Q(t) = ft - t^2$. At what time, both the buses have same velocity ?

- (A) $\frac{\alpha - f}{1 + \beta}$ (B) $\frac{\alpha + f}{2(\beta - 1)}$
(C) $\frac{\alpha + f}{2(1 + \beta)}$ (D) $\frac{f - \alpha}{2(1 + \beta)}$

Space for your notes:

Q4 - 26 June - Shift 1

A person is standing in an elevator. In which situation, he experiences weight loss ?

- (A) When the elevator moves upward with constant acceleration
(B) When the elevator moves downward with constant acceleration
(C) When the elevator moves upward with uniform velocity
(D) When the elevator moves downward with uniform velocity

Space for your notes:

Q5 - 26 June - Shift 1

An object is thrown vertically upwards. At its maximum height, which of the following quantity becomes zero ?

- (A) Momentum (B) Potential energy
(C) Acceleration (D) Force

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Q6 - 26 June - Shift 1

A ball of mass 0.5 kg is dropped from the height of 10m. The height, at which the magnitude of velocity becomes equal to the magnitude of acceleration due to gravity, is m. (Use $g = 10 \text{ m/s}^2$).

*Space for your notes:***Q7 - 26 June - Shift 2**

A ball is projected vertically upward with an initial velocity of 50 ms^{-1} at $t = 0\text{s}$. At $t = 2\text{s}$, another ball is projected vertically upward with same velocity.

Space for your notes:

At $t = \underline{\hspace{2cm}}$ s, second ball will meet the first ball ($g = 10 \text{ ms}^{-2}$).

Q8 - 28 June - Shift 1

A man of 60 kg is running on the road and suddenly jumps into a stationary trolley car of mass 120 kg. Then, the trolley car starts moving with velocity 2 ms^{-1} . The velocity of the running man was $\underline{\hspace{2cm}} \text{ ms}^{-1}$, when he jumps into the car.

*Space for your notes:***Q9 - 28 June - Shift 2**

Questions

MathonGo

Velocity (v) and acceleration (a) in two systems of

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units 1 and 2 are related as $v_2 = \frac{n}{m} v_1$ and

$a_2 = \frac{a_1}{mn}$ respectively. Here m and n are

constants. The relations for distance and time in

two systems respectively are:

(A) $\frac{n^3}{m^3} L_1 = L_2$ and $\frac{n^2}{m} T_1 = T_2$

(B) $L_1 = \frac{n^4}{m^2} L_2$ and $T_1 = \frac{n^2}{m} T_2$

(C) $L_1 = \frac{n^2}{m} L_2$ and $T_1 = \frac{n^4}{m^2} T_2$

(D) $\frac{n^2}{m} L_1 = L_2$ and $\frac{n^4}{m^2} T_1 = T_2$

Q10 - 28 June - Shift 2

A car covers AB distance with first one-third at

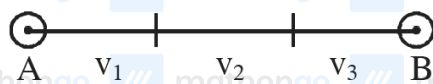
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velocity $v_1 \text{ ms}^{-1}$, second one-third at $v_2 \text{ ms}^{-1}$ and

last one-third at $v_3 \text{ ms}^{-1}$. If $v_3 = 3v_1$, $v_2 = 2v_1$ and

$v_1 = 11 \text{ ms}^{-1}$ then the average velocity of the car is

_____ ms^{-1} .



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Questions

MathonGo

Q11 - 29 June - Shift 1

Two balls A and B are placed at the top of 180 m tall tower. Ball A is released from the top at $t = 0$ s. Ball B is thrown vertically down with an initial velocity 'u' at $t = 2$ s. After a certain time, both balls meet 100 m above the ground. Find the value of 'u' in ms^{-1} . [use $g = 10 \text{ ms}^{-2}$]

- (A) 10 (B) 15
(C) 20 (D) 30

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Q12 - 29 June - Shift 2

A small toy starts moving from the position of rest under a constant acceleration. If it travels a distance of 10m in t s, the distance travelled by the toy in the next t s will be :

- (A) 10m (B) 20m
(C) 30m (D) 40m

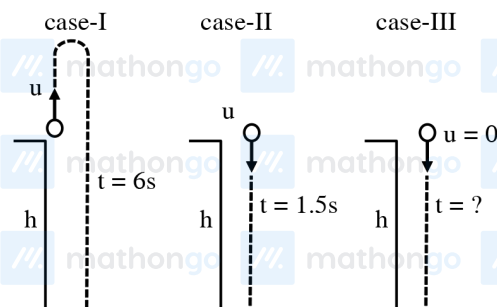
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Q12 (C)

Q1 (3)

Let height of tower be h and speed of projection in first two cases be u .



For case-I : 2nd equation $s = ut + \frac{1}{2}at^2$
 $h = -u(6) + \frac{1}{2}g(6)^2$

$$h = -6u + 18g \dots (i)$$

For case-II : $h = u(1.5) + \frac{1}{2}g(1.5)^2$

$$h = 1.5u + \frac{2.25g}{2} \dots (ii)$$

Multiplying equation (ii) by 4 we get

$$4h = 6u + 4.5g \dots (iii)$$

equation (i) + equation (iii) we get $5h = 22.5g$

$$h = 4.5g \dots (iv)$$

For case-III :

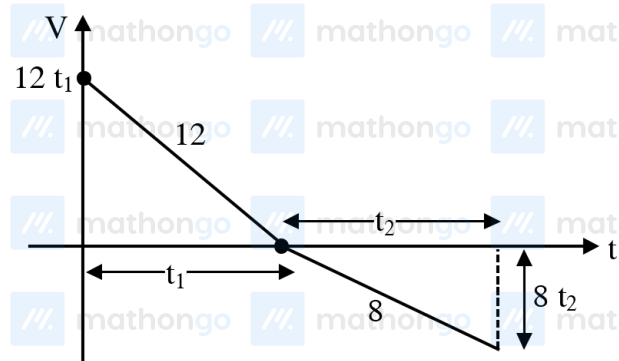
$$h = 0 + \frac{1}{2}gt^2 \dots (v)$$

Using equation (4) & equation (5)

$$4.5g = \frac{1}{2}gt^2$$

$$t^2 = 9 \Rightarrow t = 3s$$

Q2 (B)



$$6t_1^2 = 4t_2^2$$

Q3 (D)

$$X_P(t) = \alpha t + \beta t^2 \quad X_Q = ft - t^2$$

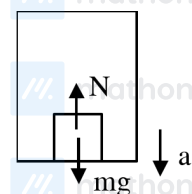
$$V_P(t) = \alpha + 2\beta t \quad V_Q = f - 2t$$

$$V_P = V_Q$$

$$\alpha + 2\beta t = f - 2t$$

$$t = \frac{f - \alpha}{2\beta + 2}$$

Q4 (B)



$$mg - N = ma$$

$$\Rightarrow N = m(g - a)$$

$\therefore$ Person experiences weightloss, when acceleration of lift is downward.

Q5 (A)

At maximum height, $V = 0$

$\therefore$ Momentum of object is zero.

Q6 (5)

$$v^2 = u^2 + 2as$$

$$100 = 0 + 2(10)s$$

$$S = 5\text{m}$$

$$\text{Height from ground} = 10 - 5 = 5\text{m}$$

Q7 (6)

Let they meet at $t = t$

So first ball gets t sec.

& 2nd gets $(t - 2)$ sec. & they will meet at same

height

$$h_1 = 50t - \frac{1}{2}gt^2$$

$$h_2 = 50(t - 2) - \frac{1}{2}g(t - 2)^2$$

$$h_1 = h_2$$

$$50t - \frac{1}{2}gt^2 = 50(t - 2) - \frac{1}{2}g(t - 2)^2$$

$$100 = \frac{1}{2}g[t^2 - (t - 2)^2]$$

$$100 = \frac{10}{2}[4t - 4]$$

$$5 = t - 1$$

$$t = 6 \text{ sec.}$$

Q8 (6)

Taking the system as man and trolley and using conservation of linear momentum.

$$60 \times v = (60 + 120) \times 2$$

$$\Rightarrow v = 6 \text{ m/s}$$

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Q9 (A)

$$\frac{L_2}{T_2} = \frac{n}{m^2} \frac{L_1}{T_1}$$

$$\frac{L_2}{T_2^2} = \frac{L_1}{T_1^2 \times mn}$$

$$\frac{n}{m^2} \times \frac{T_2}{T_1} = \frac{T_2^2}{T_1^2 \times mn}$$

$$\frac{n^2}{m} = \frac{T_2}{T_1}$$

$$\frac{L_2}{L_1} = \frac{n^4}{m^2} \times \frac{1}{mn}$$

$$\frac{L_2}{L_1} = \frac{n^3}{m^3}$$

Q10 (18)

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$$\vec{v} = \frac{\text{Displacement}}{\text{time}}$$

(Let displacement be l)

$$= \frac{l}{\left(\frac{l}{V_3} + \frac{l}{V_2} + \frac{l}{V_1}\right) \frac{1}{3}}$$

$$= \frac{3}{\frac{1}{V_1} + \frac{1}{V_2} + \frac{1}{V_3}} = \frac{3}{\frac{1}{11} + \frac{1}{22} + \frac{1}{33}}$$

$$= 18 \text{ m/s}$$

Q11 (D)

Let they meet at time t .

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 80}{10}}$$

$$= 4 \text{ sec}$$

Time taken by ball B to meet A = 2 sec

$$\text{using } S = ut + \frac{1}{2}at^2$$

$$-80 = -u \times 2 + \frac{1}{2}(-10)(2)^2$$

$$u = 30$$

Q12 (C)

$u = 0$, Say acceleration is a

For t s $10 = \frac{1}{2}at^2$

For $2t$ s $10 + x = \frac{1}{2}a(2t)^2$

$$\frac{10 + x}{10} = \frac{4}{1}$$

$$x = 30 \text{ m}$$